

Task Report

1. Graph-Level Statistics (Before Any Attacks)

Metric	Value	Explanation
Number of nodes	4039	Total users in the Facebook ego network
Number of edges	88,234	Total friendship connections
Average degree	43.69	Average number of connections per user
Median degree	25	Half of the nodes have ≤ 25 connections
Degree standard deviation	52.41	Very high $\rightarrow$ presence of hubs with extremely high degree
Average clustering coefficient	0.606	Highly clustered social network
Number of components	1	Entire graph is fully connected
Largest connected component	4039	All nodes belong to the same component
Diameter (LCC)	8	Maximum shortest-path distance between any two nodes
Average shortest path	3.69	Typical small-world behavior of social networks

2. Node Features (Before and After Bot Injection)

Node 0 - Before Bot Injection

Feature	Value
degree	347
log_degree	5.8522
clustering	0.0419
triangles	2519

core	21
avg_neighbor_deg	18.96
degree_centrality	0.0859
closeness	0.3533
betweenness	0.1473
pagerank	0.0062
eigenvector	3.31e-05
community	0
degree_to_neighbor_ratio	18.30
is_core	0

Node 0 - After Injecting 100 Bots (Before Any Attack)

Feature	Value	Change
degree	347	unchanged
avg_neighbor_deg	19.32	slight increase
closeness	0.37286	increased
betweenness	0.09067	decreased
pagerank	0.00562	small decrease
eigenvector	3.45e-05	small increase
community	0	unchanged

3. Node Features After Attacks

Node 0 – After Structural Evasion Attack

Feature	Value	Notes
degree	361	number of edges increased significantly
clustering	0.0393	similar to original (still low)
triangles	2556	slight increase
degree_centrality	0.0872	increased
closeness	0.3969	noticeable increase
betweenness	0.1079	changed significantly
eigenvector	1.68e-04	large increase → node attached to high-importance nodes
community	0	unchanged

Node 0 – After Graph Poisoning Attack

Feature	Value	Notes
degree	347	nearly restored to original
avg_neighbor_degree	19.44	slight increase
closeness	0.37346	similar to pre-attack values
betweenness	0.08620	lower than in evasion case
eigenvector	3.48e-05	almost same as original
community	5	node was reassigned to a different community

4. Classification Performance

Baseline (Before Any Attacks)

Evaluation on injected graph (100 bots):

Metric	Value
Precision	1.00
Recall	1.00
F1-score	1.00
ROC-AUC	1.00

After Structural Evasion Attack

Evaluation using baseline model on evaded graph:

Metric	Value	Interpretation
Precision	1.00	No benign users misclassified
Recall	0.79	~21% of bots successfully evaded detection
F1-score	0.88	Performance degraded due to lower recall

After Graph Poisoning Attack

Metric	Value	Interpretation
Precision	1.00	No false positives
Recall	1.00	No false negatives
F1-score	1.00	No degradation observed
ROC-AUC	1.00	Perfect separation on clean data